Course : Statistics for Data Science Professor Name: Sergiy Nokhrin Date : Nov 20 2025

Obesity Analysis using Lifestyle and Physical Condition Factors

Team Members

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Introduction and Project Motivation

Introduction

- Obesity is a major global health issue influenced by lifestyle habits and personal factors.
- Dataset of 2,000 entries on lifestyle, habits, and demographics.

	Gender	Age	Height	Weight	FamilyOverweight	HighCalFreq	VegMealFreq	MealsPerDay	Snacking	SMOKE	WaterIntake	TrackCalories	ExerciseFreq	ScreenTime	AlcoholFreq	TransportMode	ObesityLevel
0	Female	21.0	1.62	64.0	yes	no	2.0	3.0	Sometimes	no	2.0	no	0.0	1.0	no	Public_Transportation	Normal_Weight
1	Female	21.0	1.52	56.0	yes	no	3.0	3.0	Sometimes	yes	3.0	yes	3.0	0.0	Sometimes	Public_Transportation	Normal_Weight

- Our goal is to understand which factors matter the most and how they relate to overall health.

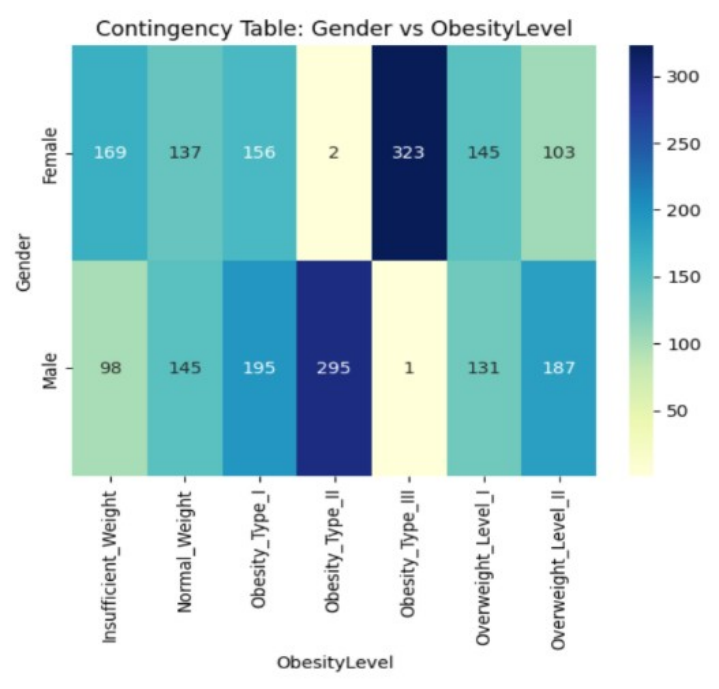
- **Data Preparation :** Renamed columns for easier analysis, converted key variables (Gender, Snacking, AlcoholFreq, TransportMode, ObesityLevel) to categorical type, and removed duplicate rows to ensure overall data quality.
- **Created New Features:** BMI

```
df['BMI'] = df['Weight'] / (df['Height']**2)
```

Chi-Square Test: Gender and Obesity Level

Hypothesis

H_0: Gender and Obesity Level are independent (no association).
H_1: Gender and Obesity Level are not independent (significant association).



Chi-Square Test Results:

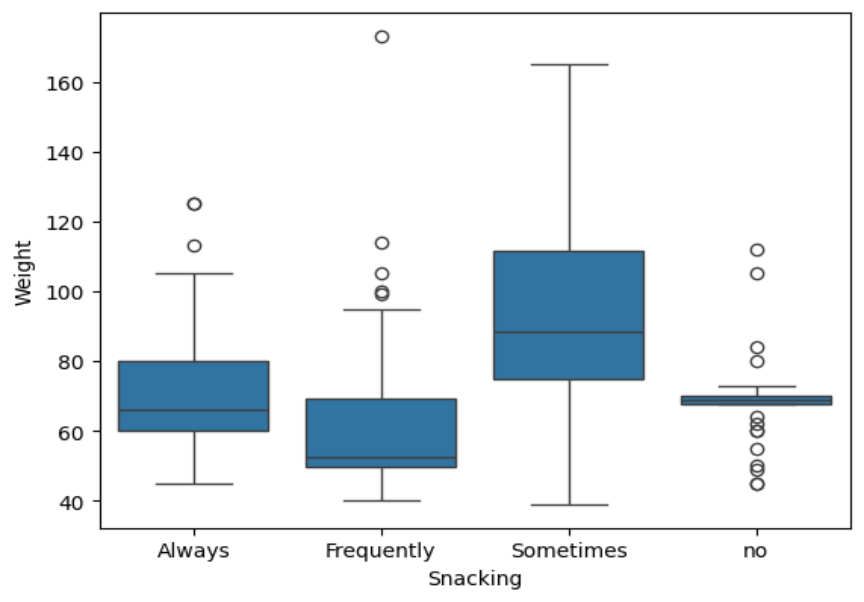
- Chi-square Statistic: 657.45
- Degrees of Freedom: 6
- P-value: 0.0000

Conclusion:
Since the p-value is 0.0000, there is a **significant association** between Gender and Obesity Level. This means obesity levels differ meaningfully between males and females.

ANOVA: Snacking vs Weight

Hypothesis

- **H_0:** Mean weight is equal across all snacking groups.
- **H_1:** At least one group mean is different.



Test Multiple Comparison ttest_ind FWER=0.05
method=bonf alphacSidak=0.01, alphacBonf=0.008

group1	group2	stat	pval	pval_corr	reject
Always	Frequently	4.6444	0.0	0.0	True
Always	Sometimes	-5.8805	0.0	0.0	True
Always	no	0.7432	0.4594	1.0	False
Frequently	Sometimes	-19.3069	0.0	0.0	True
Frequently	no	-3.2798	0.0012	0.007	True
Sometimes	no	5.5716	0.0	0.0	True

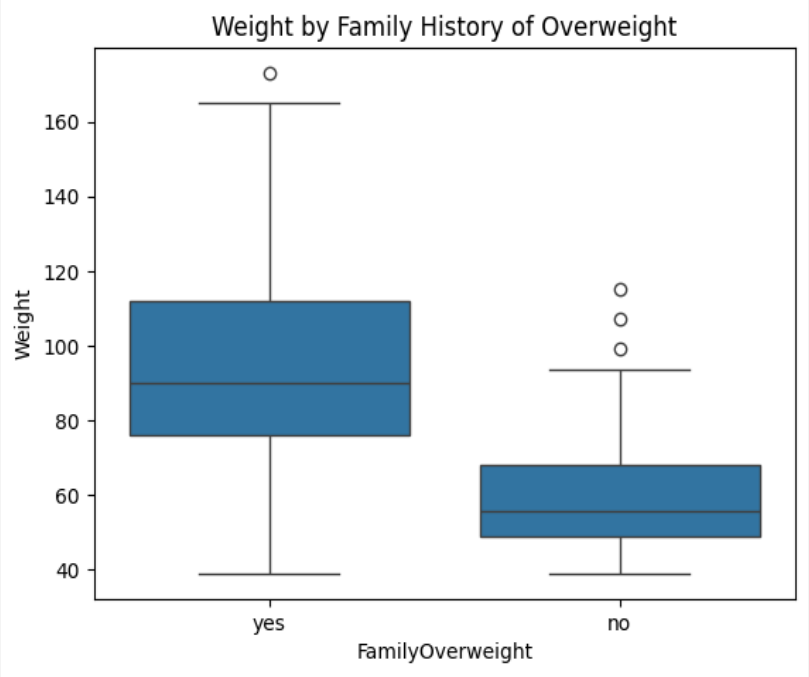
Results

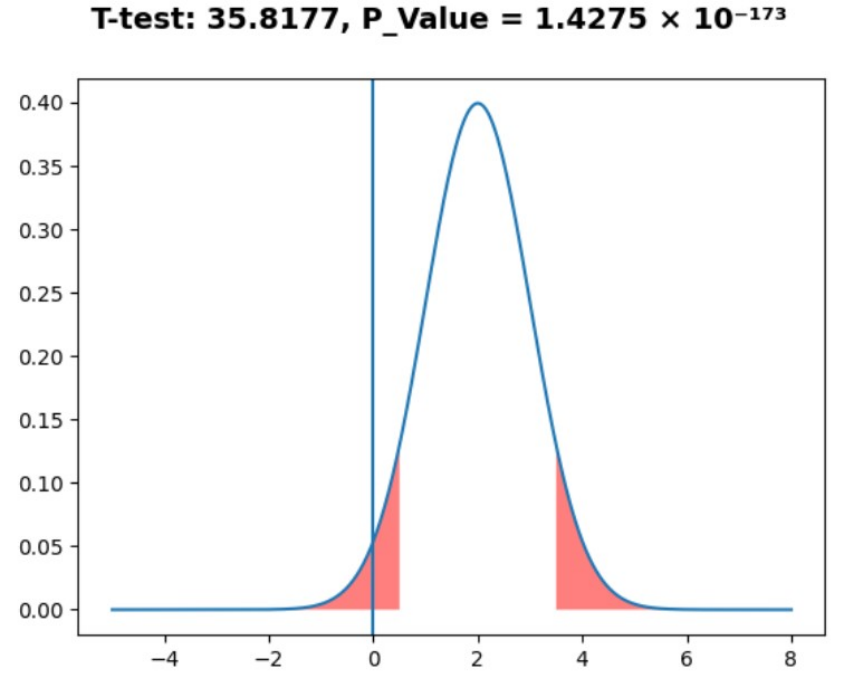
- **F-statistic:** 142.85
- **p-value:** 3.58×10^{-84}

T-Test: Weight and Family History of Overweight

Hypothesis

- **H_0 :** There is no difference in average weight between the two groups.
- **H_1 :** There *is* a significant difference in average weight between the two groups.

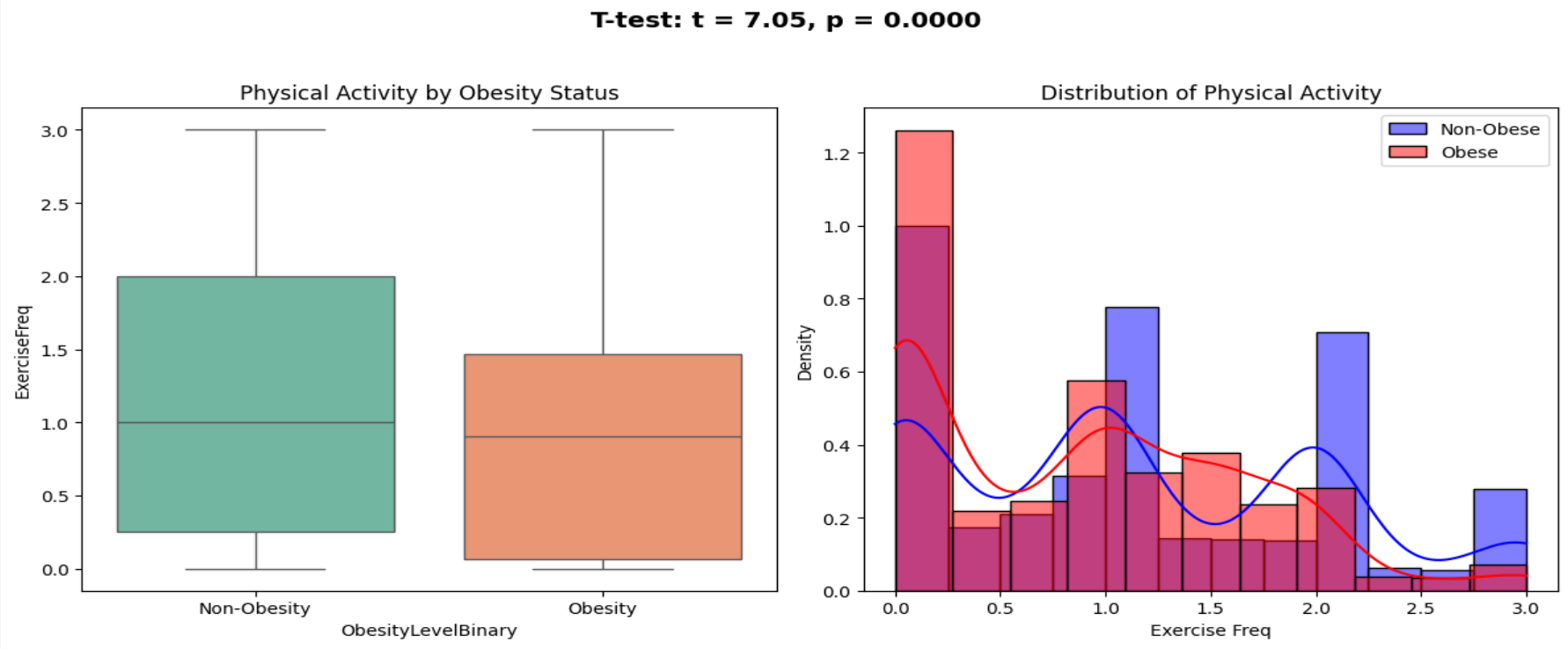




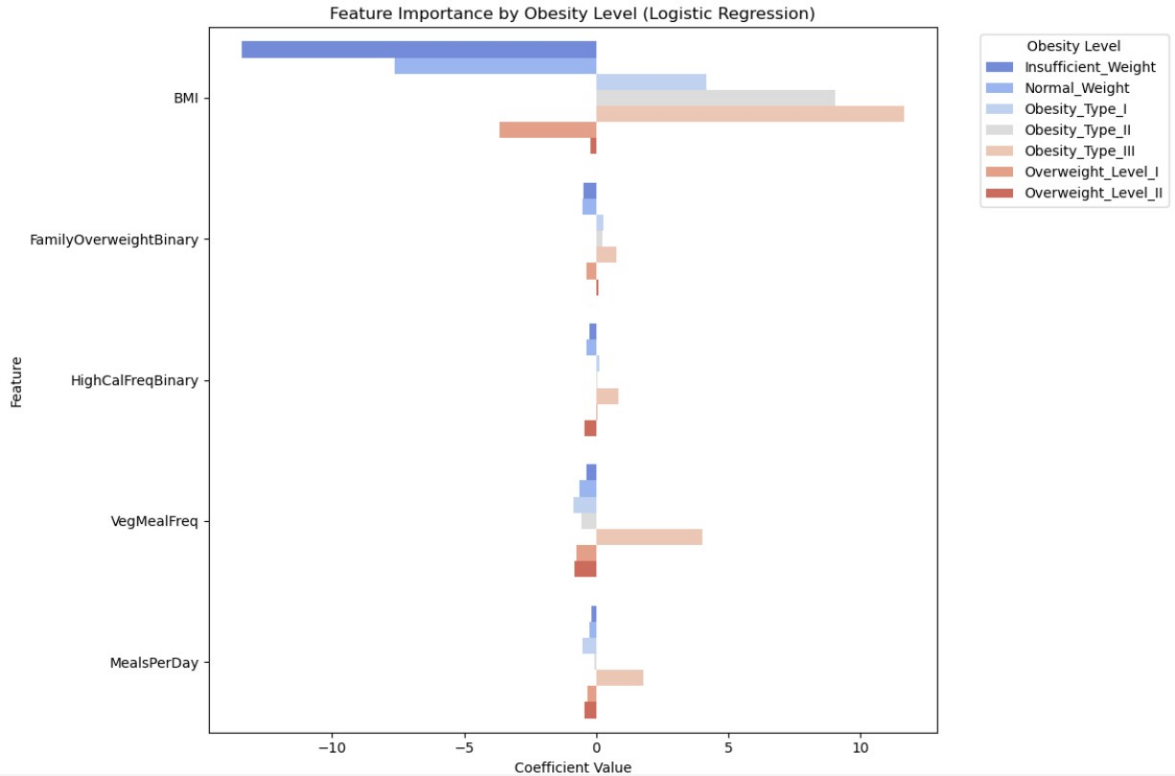
T-Test: Exercise Frequency vs Obesity Status

Hypothesis

- **H_0 :** Obese and non-obese people exercise about the same amount.
- **H_1 :** Non-obese people tend to exercise more than obese people.



Logistic Regression Model for Obesity Prediction



Positive impact: BMI, High-Calorie food Intake, Smoke

Negative impact: Tracking Calories

Negligible impact: Age, Screen Time, Water intake & meals per day.

Model Version	Accuracy
Model 1: With BMI	0.995
Model 2: Without BMI	0.683
Model 3: With Weight Only	0.880

```
df['BMI'] = df['Weight'] / (df['Height']**2)
```

Conclusion

- Physical factors especially **BMI and weight**, are the most powerful indicators of obesity.
- Lifestyle habits such as exercise, snacking, and family history show meaningful differences across groups but have a **weaker predictive impact**.
- Statistical tests confirmed significant relationships between obesity and behaviours, but **logistic regression showed that lifestyle factors alone cannot accurately predict obesity**.
- Overall, a **data-driven approach** combining physical measures with lifestyle insights provides the best understanding of obesity patterns and risk.